Time Series Is Worth 64 Words: Long-Term Forecasting With Transformer

Murphy Dao Klein, Kai Emilio Gangi, Javohir Abdurazzakov, Yung Hsueh Lee

CS 4782

Introduction

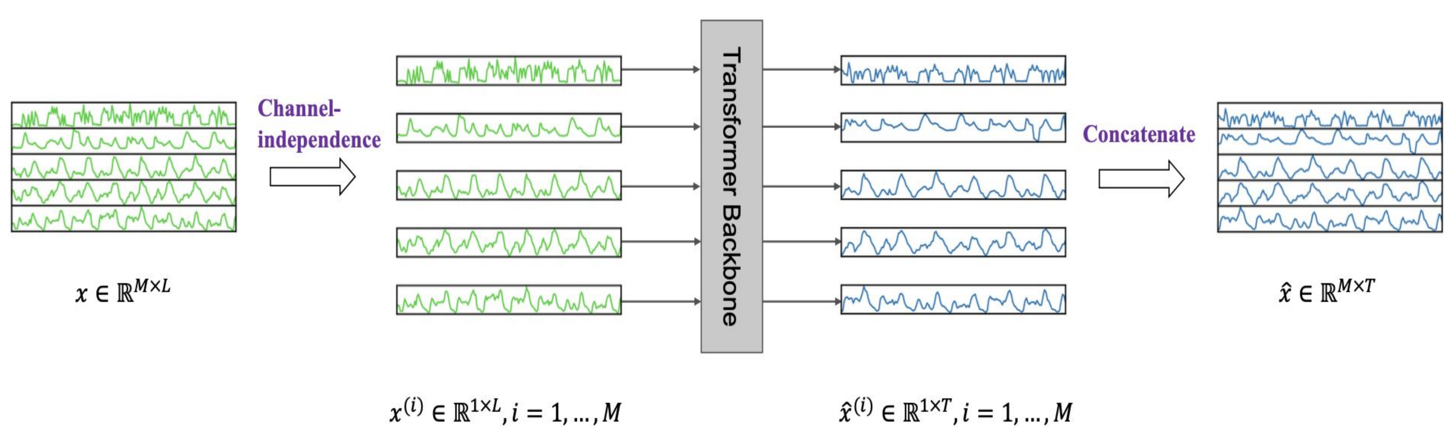
Background:

- Time series are difficult
- Previous time series models use single points, while transformers are very expensive
- Leverage ViT's to improve time series forecasting

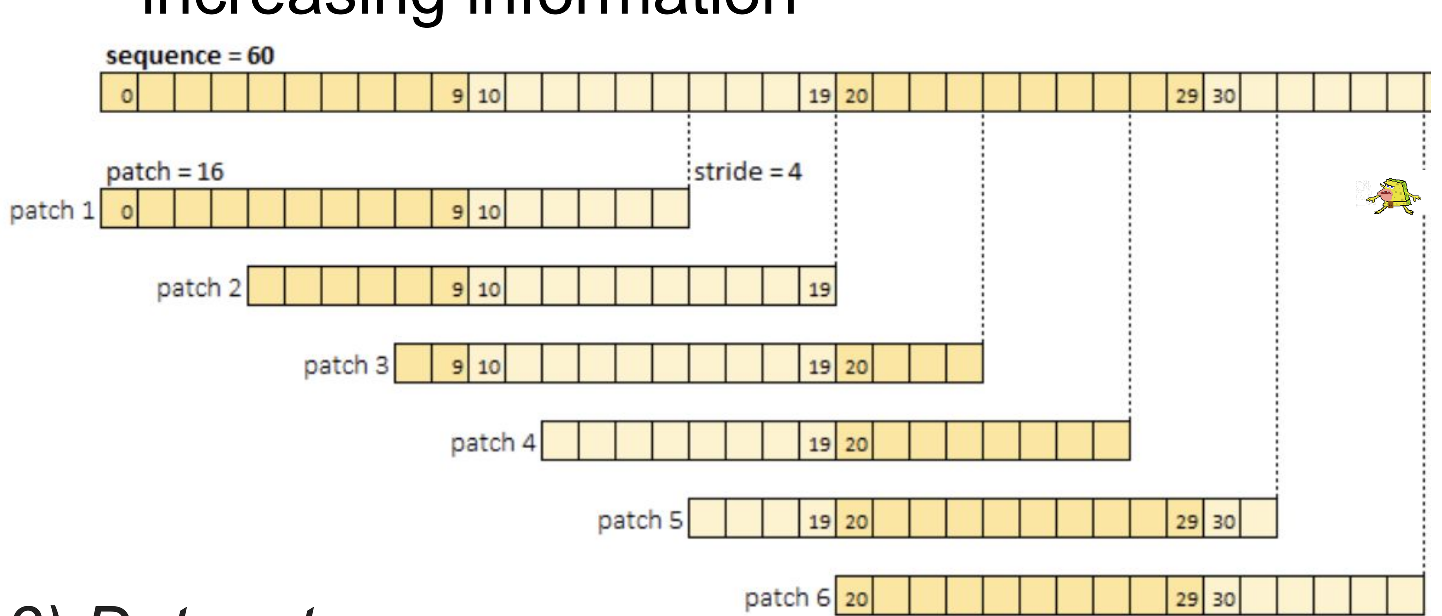
Model Formulation

Objectives:

1) *Channel Independence*
   - Reduces overfitting
   - Gives more data to use
   - Simpler
   - Inductive bias that time series share patterns



2) *Patching*
   - Reduces computational complexity
   - Enhances locality
   - Makes more sense as a transformer token
   - Enable to use longer lookback window, increasing information

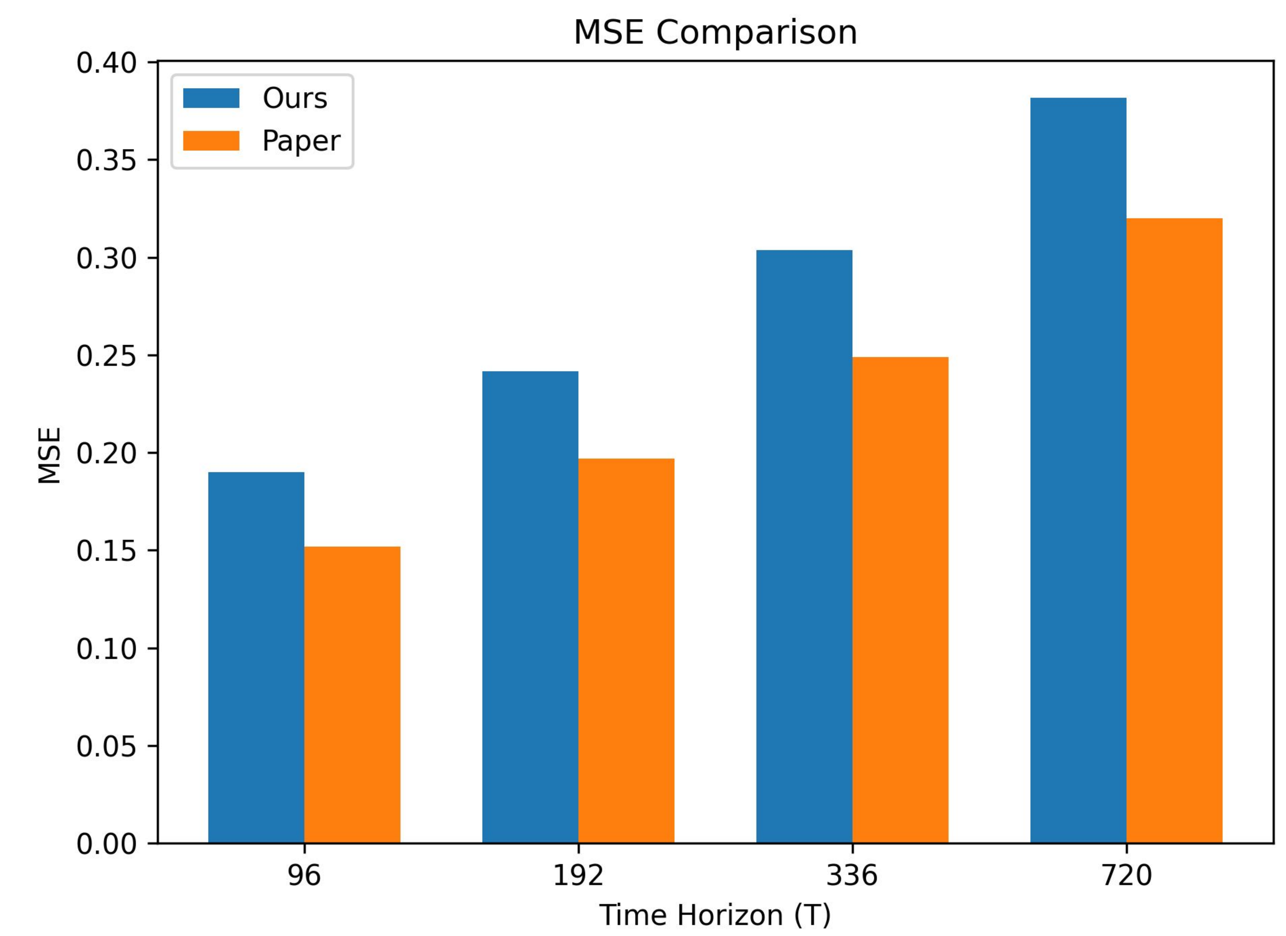


3) *Dataset usage*
   - Merge weather dataset for past 5 years
   - Followed 70/20/10 split

Results

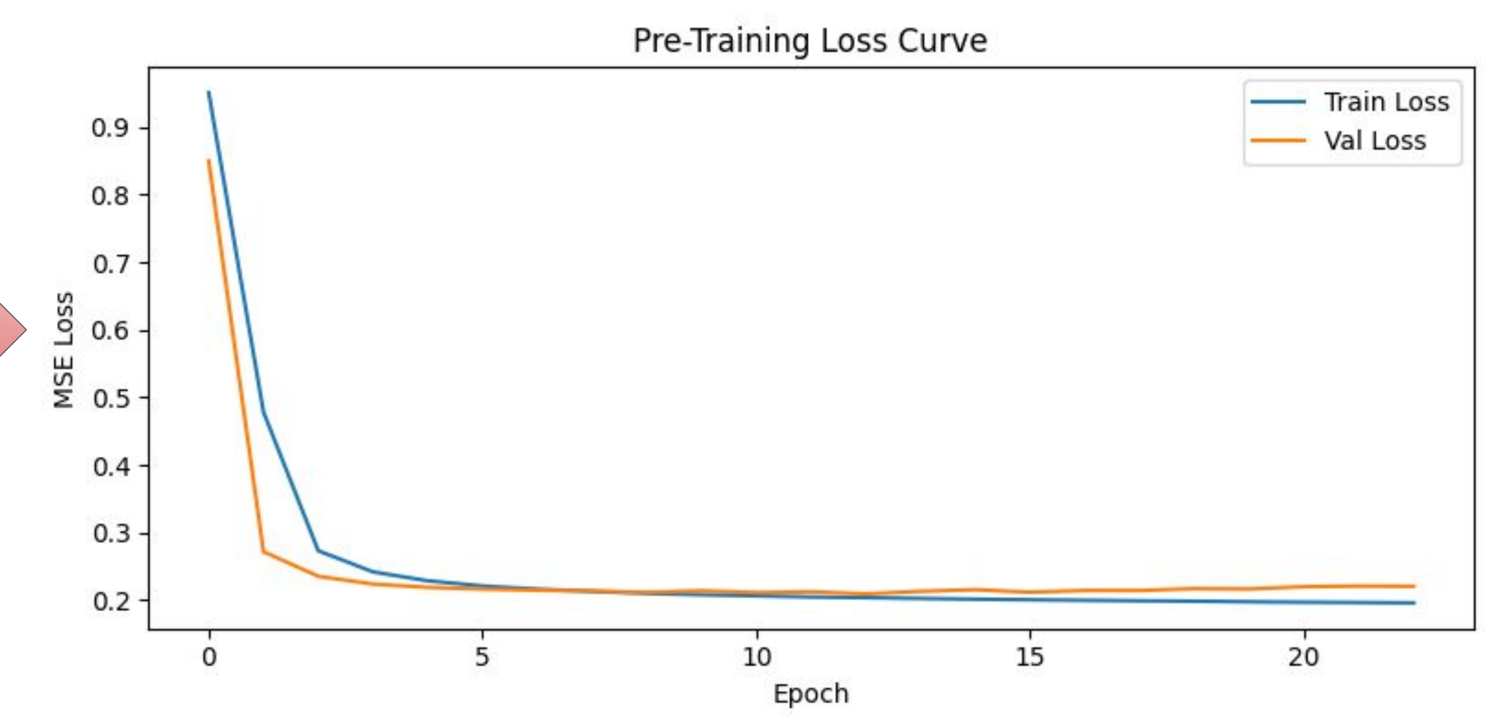
Supervised Results

- Input a sequence of patches
- Output time window prediction directly



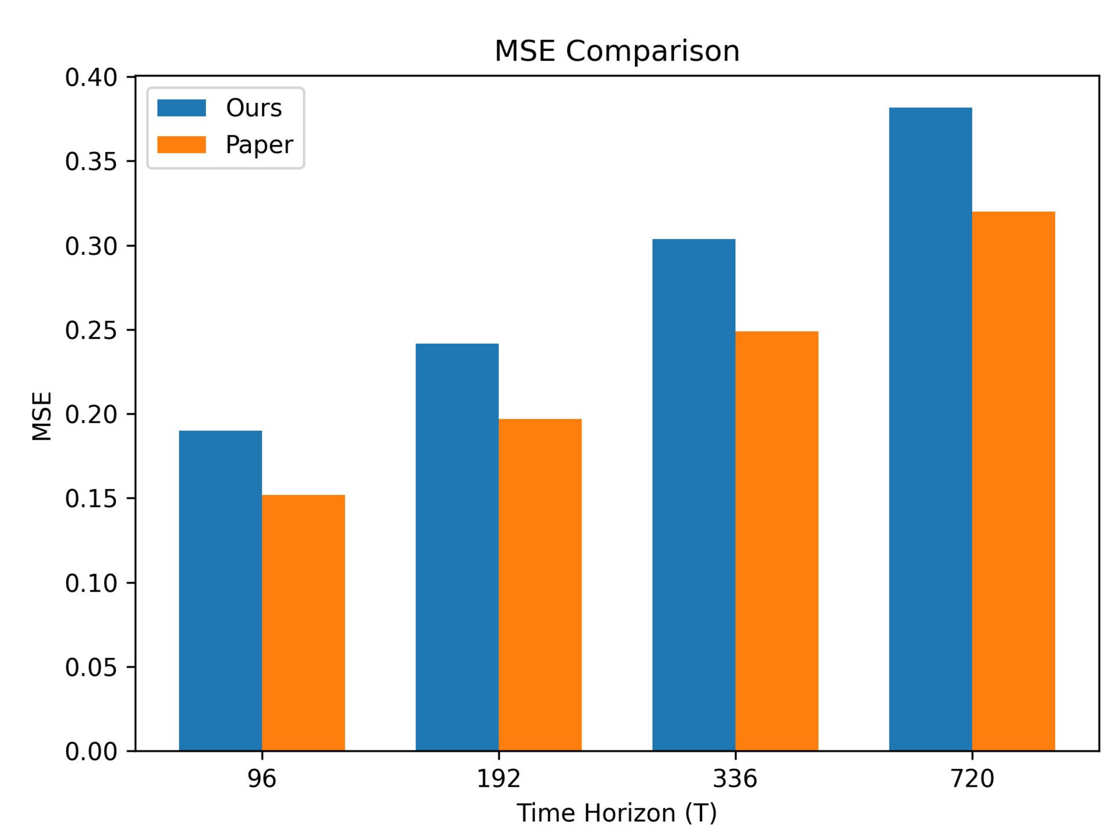
Pre-Training Results

- Self-supervised rep. learning
- Masked patch reconstruction



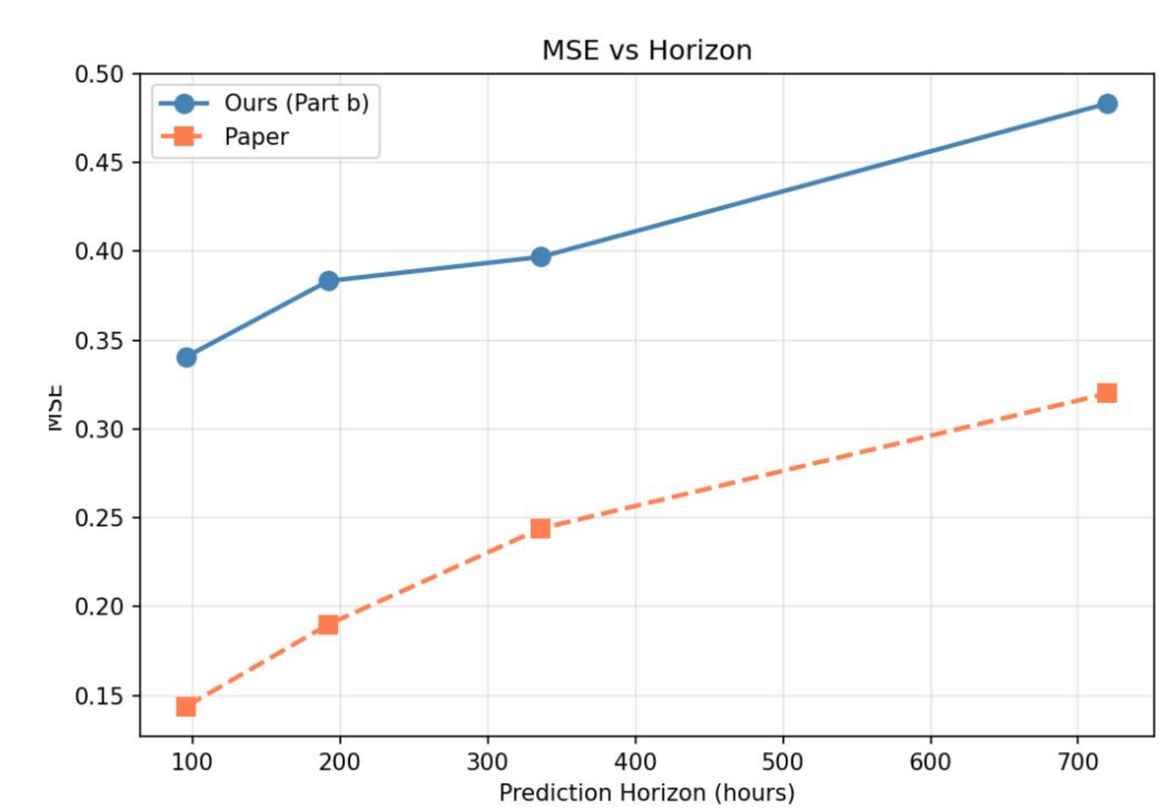
Linear Probing Results

- Attach small NN head to pre-trained model
- Freeze pre-trained model, only train the head



Fine-Tuning Results

- Start with linear probing
- End with end-to-end fine tuning.



Future Work

- MSE higher than the paper. Maybe early stopping prevents double descent?
- Bigger dataset, more compute power
- Reduce overlapping patches to improve computation efficiency